Rings and modular arithmetic

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https://github.com/xinhi6716/computer-science-self-study

14.1 Ring axioms

A nonempty set R, equipped with addition and multiplication, is a **ring** if:

1. $(R, +)$ is an abelian group. Addition is closed, associative and commutative, has identity 0, and gives every element an additive inverse.
2. Multiplication is closed and associative.
3. Both distributive laws hold, $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

Following the convention for this chapter, a general ring need not have commutative multiplication or a multiplicative identity. A **ring with identity** has an element 1; a **commutative ring** also satisfies $ab = ba$. Assumptions such as commutativity and the existence of 1 will be stated when discussing modular arithmetic, units, integral domains, fields and CRT. Properties of these special rings cannot silently be used for all rings.

If a ring with identity satisfies $0 = 1$, then every $a = a1 = a0 = 0$. It has just one element and is the **zero ring**. Integral domains and fields exclude it.

14.1 Ring axioms (continued)

The distinction between a subring and an ideal concerns absorption under multiplication. Containing 0 and being closed is not enough to make a subset an ideal.

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14.2 Examples of rings

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are commutative rings with identity.
- $M_n(\mathbb{R})$, with ordinary matrix addition and multiplication, is a ring with identity. For $n \geq 2$, multiplication is generally not commutative, so steps requiring commutativity need justification.
- The residue classes in $\mathbb{Z}_n$ form a ring under addition and multiplication followed by reduction modulo n.
- Polynomial rings $F[x]$ will be used later; their coefficients come from a field or ring.

Ring elements need not be numbers. Functions also form rings under pointwise addition and multiplication.

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Two explicit counterexamples in a matrix ring

Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus $AB \neq BA$. Also $A \neq 0$ but $A^2 = 0$, so A is a nonzero zero divisor. In general, $AX = AY$ does not allow cancellation of A. Matrix multiplication still satisfies associativity and both distributive laws. Noncommutativity does not prevent a structure from being a ring.

A ring whose elements are sets

For a finite set U, take the subsets of U as elements, symmetric difference $\triangle$ as addition, and intersection $\cap$ as multiplication. The empty set is the additive identity. Since $A \triangle A = \varnothing$, each A is its own additive inverse. The multiplicative identity is U.

For each $u \in U$, use a bit to record membership. Symmetric difference becomes bitwise XOR, and intersection becomes bitwise AND. Each ring axiom holds coordinate by coordinate by the corresponding axiom in $\mathbb{Z}_2$. This is a commutative ring. If $|U| = k$, it has 2^k elements and is isomorphic to $(\mathbb{Z}_2)^k$.

If $k \geq 2$, two distinct singletons A, B satisfy $A \cap B = \varnothing$. They are nonzero zero divisors, so the ring is not a field. Yet every element satisfies $A^2 = A$, the **idempotent** property. Whether division is possible depends on the operations, not on whether the elements look like numbers, sets, matrices or functions.

14.3 Basic consequences of the axioms

For all $a, b \in R$,

$$a0 = 0 = 0a, \qquad a(-b) = -(ab) = (-a)b.$$

Indeed, $a0 = a(0 + 0) = a0 + a0$; cancellation in the additive group gives $a0 = 0$. The argument uses distributivity and additive inverses, rather than assuming the familiar integer rule.

The additive identity is unique, because $0 + 0'$ equals both 0 and $0'$. If b, c are additive inverses of a, then

$$b = b + (a + c) = (b + a) + c = c.$$

Thus additive inverses are unique. Since $ab + a(-b) = a(b - b) = 0$, uniqueness gives $a(-b) = -(ab)$. The other side is analogous, and $(-a)(-b) = ab$ follows. None of these results requires commutative multiplication.

If $ab = 0$ with **both** $a \neq 0$ and $b \neq 0$, then a, b are nonzero zero divisors. Knowing only $b \neq 0$, while allowing $a = 0$, does not establish that b is a zero divisor.

14.3 Basic consequences of the axioms (continued)

In $\mathbb{Z}_6$, $2 \cdot 3 = 0$, so multiplicative cancellation does not hold universally.

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14.4 Units, zero divisors and integral domains

In a commutative ring with identity, u is a **unit** if some v satisfies $uv = 1$. The units form a multiplicative group $R^\times$. In a noncommutative ring, both $uv = 1$ and $vu = 1$ are required. An **integral domain** is a nonzero commutative ring with identity satisfying

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

The integers form an integral domain. For composite $n > 1$, $\mathbb{Z}_n$ does not.

Proof of the prime-modulus criterion. If p is prime and $ab \equiv 0 \pmod{p}$, Euclid's lemma gives $p \mid a$ or $p \mid b$. Conversely, if $n = rs$ with $1 < r, s < n$, the classes of r, s are nonzero and have product zero.

An integral domain need not be a field. In $\mathbb{Z}$, only $1, -1$ have multiplicative inverses, although other nonzero elements exist.

A **field** is a nonzero commutative ring with identity in which every nonzero element is a unit. Division by nonzero elements is available in a field. A general ring guarantees additive inverses, but not multiplicative inverses.

Cancellation and finite integral domains

In a commutative ring, $a \neq 0$ and $ab = ac$ yield only $a(b - c) = 0$. If a is not a zero divisor, this implies $b = c$. Conversely, cancellation for a particular nonzero a forces $b = 0$ whenever $ab = 0 = a0$. Thus cancellation by a is equivalent to a not being a zero divisor.

In a nonzero commutative ring with identity, a nonzero unit cannot be a zero divisor. Multiplying $ab = 0$ by a^{-1} gives $b = 0$. Every field is therefore an integral domain. The converse fails for $\mathbb{Z}$, where 2 is not invertible.

Every finite integral domain is a field. For $a \neq 0$, the map $m_a : x \mapsto ax$ is injective by cancellation. An injection from a finite set to itself is surjective, so $1 = ax$ for some x. Every nonzero a is invertible. Finiteness is essential. On $\mathbb{Z}$, $x \mapsto 2x$ is injective but not surjective.

The unit group and characteristic

In a ring with identity, v is the multiplicative inverse of u if $uv = vu = 1$. In a commutative ring, one equality suffices. The inverse is unique, since $v = v(uw) = (vu)w = w$. Products of units are units, with

$$(uv)^{-1} = v^{-1}u^{-1}.$$

Consequently all units form a group. In a noncommutative ring, the order of these two inverse factors cannot be reversed without justification.

The **characteristic** is the least positive integer p such that $p \cdot 1 = 0$, or 0 if no such integer exists. An integral domain has $1 \neq 0$, so its characteristic is not 1. If a positive characteristic were composite, $p = ab$ with $1 < a, b < p$, then

$$(a \cdot 1)(b \cdot 1) = 0.$$

Minimality of p makes both factors nonzero, a contradiction. A positive characteristic of an integral domain must therefore be prime. In a finite integral domain, the sequence $0, 1, 2 \cdot 1, \ldots$ repeats, so its characteristic is a prime.

The unit group and characteristic (continued)

For example, $\mathbb{Z}_{12}$ has characteristic 12 and is not an integral domain. The characteristic of $\mathbb{Q}$ is 0. Characteristic measures repeated addition of the identity, rather than the numeral used to name it.

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14.5 Subrings and ideals

A nonempty subset $S \subseteq R$ is a **subring** if it is itself a ring under the restricted operations. Our convention does not require S to contain 1_R. The exact test is

$$a, b \in S \implies a - b \in S \text{ and } ab \in S.$$

For sufficiency, taking $a = b$ gives $0 \in S$, then $0 - a$ gives additive inverses and $a - (-b)$ gives additive closure. Associativity and distributivity are inherited. Necessity follows directly from the subring axioms.

A **unital subring** must additionally contain 1_R. For example, $2\mathbb{Z}$ is a subring and ideal of $\mathbb{Z}$, but does not contain 1 and has no multiplicative identity of its own under the inherited multiplication. Do not change subring conventions midway through an argument.

A **two-sided ideal** I is an additive subgroup satisfying $ri, ir \in I$ for all $r \in R, i \in I$. This is stronger than multiplication being closed only within I. If an ideal in a ring with identity contains 1, then $r = r1 \in I$ for every r, so $I = R$.

Define operations on cosets by

14.5 Subrings and ideals (continued)

$$(a + I) + (b + I) = (a + b) + I, \qquad (a + I)(b + I) = ab + I.$$

Replacing representatives by $a + i, b + j$ changes the product by $aj + ib + ij$. Two-sided absorption puts every term in I; the change in the sum also lies in I. Both operations are well defined, producing the **quotient ring** R/I. In a noncommutative ring, ib cannot arbitrarily become bi. An arbitrary subring lacks the absorption needed to descend multiplication to cosets.

For instance, the constants $\mathbb{Z} \subseteq \mathbb{Z}[x]$ form a subring but not an ideal, since multiplying an integer by x need not give a constant.

Finite subrings and the limits of ideal closure

A nonempty finite subset S closed under addition and multiplication is already a subring. For $a \in S$, some positive multiples satisfy $ia = ja$, with $i < j$. Additive cancellation in R yields $(j-i)a = 0$. This positive multiple lies in S, so $0 \in S$, and

$$-a = (j-i-1)a \in S.$$

If the coefficient is zero, use the zero element just established. Additive inverses and closure make S an additive subgroup.

Finiteness cannot be omitted. Nonnegative integers are closed under ordinary addition and multiplication and contain $0, 1$, but have no additive inverse for 1. They are not a subring. Also, $\mathbb{Z}$ is a unital subring of $\mathbb{Q}$, but not an ideal because $\frac{1}{2} \cdot 1 \notin \mathbb{Z}$.

A nonzero commutative ring with identity having exactly the ideals $\{0\}, R$ is a field. For $a \neq 0$, the principal ideal $(a) = \{ra : r \in R\}$ is nonzero, hence equals R; thus $1 = ra$ and a is invertible. Conversely, any nonzero ideal of a field contains $a \neq 0$; multiplying by a^{-1} puts 1 in the ideal, so it is the whole field. The nonzero hypothesis keeps the two ideals distinct.

Ideals of the integers and of residue rings

Every ideal $I \subseteq \mathbb{Z}$ equals $m\mathbb{Z}$ for a unique $m \geq 0$. If $I = \{0\}$, take $m = 0$. Otherwise take the least positive element m of I. For $a \in I$, division gives $a = qm + r$, $0 \leq r < m$. Since $a, qm \in I$, also $r \in I$. Minimality forces $r = 0$, so $I \subseteq m\mathbb{Z}$. Absorption gives the reverse inclusion.

For an ideal J of $\mathbb{Z}_n$, its preimage

$$I = \{a \in \mathbb{Z} : [a]_n \in J\}$$

is an ideal of $\mathbb{Z}$ containing $n\mathbb{Z}$. Thus $I = d\mathbb{Z}$ with $d \mid n$. Conversely each positive divisor $d \mid n$ gives $J = ([d])$. This is a bijection between ideals of $\mathbb{Z}_n$ and positive divisors of n. Such J has n/d elements; its quotient has d elements and is isomorphic to $\mathbb{Z}_d$.

For example, in $\mathbb{Z}_{12}$, $([4]) = \{0, 4, 8\}$. The quotient has four classes,

$$\{0,4,8\},\ \{1,5,9\},\ \{2,6,10\},\ \{3,7,11\}.$$

Sending each class to its residue modulo 4 is a bijection preserving both operations, so $\mathbb{Z}_{12}/([4]) \cong \mathbb{Z}_4$.

14.6 Operations in $\mathbb{Z}/n\mathbb{Z}$

Representatives a and $a+kn$ must determine the same class. The quotient operations are well defined because

$$(a+kn)+(b+\ell n)\equiv a+b \pmod{n},$$
$$(a+kn)(b+\ell n)\equiv ab \pmod{n}.$$

Every extra term is a multiple of n. Moreover,

$$[a] \text{ is a unit} \iff \gcd(a,n)=1.$$

Proof. If the gcd is 1, Bézout gives $sa+tn=1$, hence $[s][a]=[1]$. Conversely, if $ab+kn=1$, any common divisor of a, n divides 1.

The additive identity is $[0]$ and the multiplicative identity is $[1]$. For $n=1$, these are the same class, producing the zero ring.

Complete classification of units and zero divisors

Let $n>1$ and $[a]\neq[0]$, and put $d=\gcd(a,n)$. If $d=1$, $[a]$ is a unit. If $d>1$, set $b=n/d$. Then $1\leq b<n$ and $n\mid ab$, so $[b]\neq[0]$ and $[a][b]=0$. This explicitly constructs a zero-divisor witness. A unit cannot have zero product with a nonzero class. Thus every nonzero class belongs to exactly one of these two categories.

Euler's function $\varphi(n)$ counts the residue classes coprime to n. If the distinct prime factors are $p_1,\dots,p_r$, inclusion–exclusion on their multiples among $0,\dots,n-1$ gives

$$\varphi(n)=n\prod_{p\mid n}\left(1-\frac{1}{p}\right).$$

There are $n-1-\varphi(n)$ nonzero zero divisors. For $n=18$,

$$\varphi(18)=18(1-1/2)(1-1/3)=6.$$

The units are $1,5,7,11,13,17$; the other 11 nonzero classes are zero divisors. For example, $6\cdot 3\equiv 0$ and $5\cdot 11\equiv 1$.

Complete classification of units and zero divisors (continued)

For $n = 1$, the only class is both 0 and 1, and is invertible by the definition of a unit. This ring is neither a field nor an integral domain. The counting formula above is explicitly restricted to $n > 1$.

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Linear congruences require a solvability test

To solve $ax \equiv b \pmod{n}$, with $n \geq 1$, put $d = \gcd(a, n)$. A solution implies $n \mid ax - b$, hence $d \mid b$. Conversely, if $d \mid b$, division by d gives

$$\frac{a}{d}x \equiv \frac{b}{d} \pmod{n/d}.$$

The new coefficient is coprime to the new modulus, so there is a unique x_0 modulo n/d. All solutions modulo n are

$$x = x_0 + k(n/d), \qquad k = 0, \ldots, d - 1,$$

which are d distinct classes. If $n/d = 1$, the reduced equation is just $0 \equiv 0$, and all classes modulo n solve the original equation; no ordinary division needs to be defined.

For $6x \equiv 8 \pmod{14}$, $d = 2$ divides 8. Reduction gives $3x \equiv 4 \pmod{7}$. The inverse of 3 is 5, so $x \equiv 20 \equiv 6 \pmod{7}$. The two solutions modulo 14 are $6, 13$. Substitution gives $36 \equiv 8$ and $78 \equiv 8$. Replacing the right side by 9 gives no solution, because $2 \nmid 9$.

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14.7 Modular inverses and Euler's theorem

If $\gcd(a, n) = 1$, the extended Euclidean algorithm produces s with $sa \equiv 1 \pmod{n}$. Euler's theorem states that

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

for a coprime to n. Fermat's little theorem is the prime-modulus case, where $n = p$ is prime and $p \nmid a$:

$$a^{p-1} \equiv 1 \pmod{p}.$$

If $\gcd(a, n) > 1$ and $n > 1$, then a has no multiplicative inverse modulo n. A power congruence happening to have a solution does not make a available for division.

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Euler's theorem by permutation and cancellation

Let $n > 1$, with units $u_1, \ldots, u_\varphi$. Since $[a]$ is a unit, multiplication by a permutes the units. Indeed, $au_i = au_j$ implies $u_i = u_j$ after multiplying by a^{-1}; finiteness gives a bijection. Commutativity then gives

$$a^{\varphi(n)} u_1 \cdots u_\varphi = u_1 \cdots u_\varphi.$$

The product on the right is a unit and can be cancelled by its inverse, proving the theorem. The permutation and cancellation fail without coprimality. For example, $2^{\varphi(8)} = 2^4 \equiv 0 \pmod{8}$, not 1. For prime p, $\varphi(p) = p - 1$, yielding Fermat's little theorem. The version also allowing $p \mid a$ is $a^p \equiv a$; division by a is not justified in that case.

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Binary modular exponentiation, bit by bit

To compute $a^e \bmod n$ for $n > 1, e \geq 0$, initialize $r = 1$ and scan all binary bits of e from left to right. For each bit b, first set $r \leftarrow r^2 \bmod n$, then, if $b = 1$, set $r \leftarrow ra \bmod n$. After reading a prefix of value q, the invariant is $r \equiv a^q$. Appending b changes the exponent to $2q + b$, proving the update. For $e = 0$, return $1 \bmod n$.

For $a = 5, e = 13 = (1101)_2, n = 17$:

Bit read	Prefix q	After squaring modulo 17	After multiplying by 5 if needed
Initial	0	No squaring	1
1	1	1	5
1	3	8	6
0	6	2	2
1	13	4	3

Binary modular exponentiation, bit by bit (continued)

Thus $5^{13} \equiv 3 \pmod{17}$. An implementation initialized at $r = a$ must skip the leading significant bit; starting at a and scanning every bit double-counts it. The algorithm uses $O(\log(e + 1))$ modular multiplications. This is not the same number of bit operations; each large-integer modular multiplication has its own cost.

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14.8 Chinese Remainder Theorem

For pairwise coprime positive moduli $m_1, \ldots, m_k$, the system

$$x \equiv a_i \pmod{m_i}$$

has exactly one solution modulo $N = \prod_i m_i$. To construct it, let $N_i = N/m_i$. Choose t_i with $t_i N_i \equiv 1 \pmod{m_i}$, possible since $\gcd(N_i, m_i) = 1$. Then

$$x = \sum_i a_i t_i N_i$$

satisfies every congruence. For uniqueness, the difference of two solutions is divisible by every m_i, hence by N because the moduli are pairwise coprime.

For the example $x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}$, write $x = 2 + 3t$. Then $2 + 3t \equiv 3 \pmod{5}$, so $3t \equiv 1 \pmod{5}$. Since $3^{-1} \equiv 2 \pmod{5}$, $t \equiv 2 \pmod{5}$, giving $x \equiv 8 \pmod{15}$.

Pairwise coprimality is sufficient for this form of CRT. More general versions use ideals.

CRT as an isomorphism of ring representations

For pairwise coprime moduli greater than 1, define

$$\Theta : \mathbb{Z}_N \longrightarrow \prod_i \mathbb{Z}_{m_i}, \qquad [x]_N \longmapsto ([x]_{m_1}, \ldots, [x]_{m_k}).$$

It is well defined since each $m_i \mid N$, and it preserves addition and multiplication coordinatewise. CRT existence and uniqueness state exactly that it is bijective. Thus it is a ring isomorphism. Product-ring operations are coordinatewise operations, not decimal operations on the entries of a tuple.

For $N = 12 = 3 \cdot 4$, let $e_1 = 4, e_2 = 9$. Their images are $(1, 0), (0, 1)$, and

$$e_1^2 = e_1, \quad e_2^2 = e_2, \quad e_1 e_2 = 0, \quad e_1 + e_2 = 1 \pmod{12}.$$

Hence $\Theta^{-1}(a, b) = 4a + 9b \pmod{12}$.

CRT as an isomorphism of ring representations (continued)

CRT: each (a mod 3, b mod 4) determines one x mod 12

	b=0	b=1	b=2	b=3
a=0	x=0	x=9	x=6	x=3
a=1	x=4	x=1	x=10	x=7
a=2	x=8	x=5	x=2	x=11

Green: units

x=1, 5, 7, 11

4 ↦ (1,0)

9 ↦ (0,1)

Each cell has x = 4a+9b mod 12. All 12 residues occur once; operations are coordinatewise.

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Compatibility when moduli are not coprime

For $x \equiv a \pmod{m}, x \equiv b \pmod{n}$, put $g = \gcd(m, n)$. A solution requires $g \mid (b - a)$. If that condition holds, writing $x = a + mk$ reduces the system to

$$(m/g)k \equiv (b - a)/g \pmod{n/g}.$$

The reduced coefficient and modulus are coprime. Thus k is unique modulo n/g, and x is unique modulo $\operatorname{lcm}(m, n)$. This also proves sufficiency of the compatibility condition. Uniqueness is not generally modulo mn.

For $x \equiv 5 \pmod{8}, x \equiv 9 \pmod{12}$, $g = 4$ divides the difference 4. Set $x = 5 + 8k$. Then $2k \equiv 1 \pmod{3}$, so $k \equiv 2 \pmod{3}$, and $x \equiv 21 \pmod{24}$. Indeed, $21 \bmod 8 = 5$ and $21 \bmod 12 = 9$. Replacing the second residue by 10 makes the difference 5, not divisible by 4, so there is no solution.

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Deterministic applications of modular arithmetic

A linear congruential generator uses $x_{t+1} = (ax_t + c) \bmod m$. Its finite state space guarantees eventual repetition, but not period m. For $a = 5, c = 1, m = 16, x_0 = 0$, the actual sequence is

$0 \to 1 \to 6 \to 15 \to 12 \to 13 \to 2 \to 11 \to 8 \to 9 \to 14 \to 7 \to 4 \to 5 \to 10 \to 3 \to 0$.

This seed visits all 16 states before returning, so the period is 16. Changing only a to 6 gives $0 \to 1 \to 7 \to 11 \to 3 \to 3$, with a transient followed by a cycle of length 1. Determinism and finite periodicity imply neither independent uniform random values nor cryptographic security.

The hash function $h(k) = k \bmod 13$ maps integers to 13 buckets. The keys $13, 26, 39$ all map to 0; more than 13 distinct keys force a collision. The remainder specifies the bucket, while collision handling and distribution quality are separate issues.

When an affine encoding can be decoded uniquely

On $\mathbb{Z}_n$, $E(x) = ax + b$ is a permutation if and only if $\gcd(a, n) = 1$. In the coprime case,

$$D(y) = a^{-1}(y - b)$$

is its inverse. If $d = \gcd(a, n) > 1$, the distinct classes x and $x + n/d$ have the same image, so E is not injective.

For $n = 26, a = 5, b = 7$, $5^{-1} = 21$, $E(4) = 27 \equiv 1$, and $D(1) = 21(1 - 7) = -126 \equiv 4$. With $a = 6$, both $x = 0$ and $x = 13$ map to 7; unique decoding is impossible. This is an example of algebraic invertibility, not a claim that a simple affine cipher provides secure encryption.

14.9 Ring homomorphisms

A **ring homomorphism** $\varphi : R \to S$ satisfies, for every $a, b \in R$,

$$\varphi(a+b) = \varphi(a) + \varphi(b), \qquad \varphi(ab) = \varphi(a)\varphi(b).$$

Following our convention for general rings, preservation of 1 is not assumed. A **unital ring homomorphism** additionally satisfies $\varphi(1_R) = 1_S$. A map preserving only addition and multiplication may send 1 to another idempotent, so domain units need not become codomain units.

For a bijective homomorphism, the inverse preserves both operations. Write $x = \varphi(a), y = \varphi(b)$; then $\varphi^{-1}(x+y) = a + b$, and similarly for multiplication. A bijective homomorphism is a **ring isomorphism**. Equal cardinalities guarantee only a set bijection, not preservation of operations.

For example, evaluation $F[x] \to F$, $f \mapsto f(a)$, is a unital ring homomorphism.

What a homomorphism preserves

From $\varphi(0) = \varphi(0) + \varphi(0)$, additive cancellation gives $\varphi(0) = 0$. Then $\varphi(a) + \varphi(-a) = \varphi(0)$ yields $\varphi(-a) = -\varphi(a)$. Images of additive subgroups are additive subgroups, and images of subrings are subrings.

If I is an ideal of S, then $\varphi^{-1}(I)$ is closed under subtraction. For i in this preimage and $r \in R$, $\varphi(ri) = \varphi(r)\varphi(i) \in I$ and likewise $\varphi(ir) \in I$. Thus the preimage is an ideal of R. Conversely, the image of an ideal of R is generally only an ideal of $\operatorname{Im}\varphi$. If φ is surjective onto S, any $s \in S$ is $\varphi(r)$, which proves absorption by all of S.

For the inclusion $\mathbb{Z} \to \mathbb{Q}$, the ideal $2\mathbb{Z}$ has the same image, but this is not an ideal of $\mathbb{Q}$, since $\frac{1}{2} \cdot 2 = 1 \notin 2\mathbb{Z}$. Surjectivity is essential in that direction.

If φ is surjective and R has 1, every $s = \varphi(r)$ satisfies $\varphi(1)s = s = s\varphi(1)$. Thus $\varphi(1)$ is the multiplicative identity of S. In the surjective case, identity preservation follows from the axioms.

14.10 Kernel, image and the first isomorphism theorem

The **kernel**

$$\ker \varphi = \{r \in R : \varphi(r) = 0_S\}$$

is an ideal, and the image is a subring. Moreover, φ is injective if and only if $\ker \varphi = \{0\}$. The first isomorphism theorem states

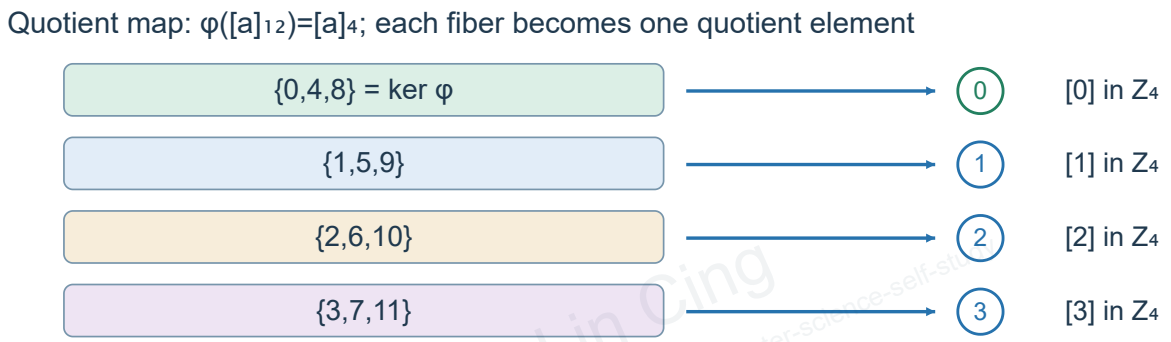
$$R/\ker \varphi \cong \operatorname{Im} \varphi.$$

For $\varphi : \mathbb{Z} \to \mathbb{Z}_n$, $k \mapsto [k]$, the kernel is $n\mathbb{Z}$, giving $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Proof. The map $r + \ker \varphi \mapsto \varphi(r)$ is well defined because representatives of one coset differ by a kernel element. It preserves both operations and surjects onto the image. Equal outputs mean the representatives differ by a kernel element, hence their cosets coincide, proving injectivity. This constructs the isomorphism rather than merely asserting a symbolic equality.

The kernel is an ideal by the preceding preimage result with $I = \{0\}$. If φ is injective, $\varphi(r) = 0 = \varphi(0)$ forces $r = 0$. Conversely, if the kernel is zero and $\varphi(a) = \varphi(b)$, then $\varphi(a - b) = 0$ forces $a = b$. Both directions of the injectivity test are proved.

Four fibers of a quotient map



The natural map $\varphi([a]_{12}) = [a]_4$ is well defined because a difference divisible by 12 is also divisible by 4. Its kernel is $\{0, 4, 8\}$. The three elements in each color group are distinct in $\mathbb{Z}_{12}$ but have one output. The quotient identifies each fiber as a single element; it does not treat the original elements in that fiber as separate quotient elements.

Changing the modulus requires well-definedness

The candidate $[a]_m \mapsto [a]_n$ is well defined if and only if $n \mid m$. Necessity follows because $[0]_m = [m]_m$ must have equal images. For sufficiency, if m divides the difference between representatives, then so does n. Once well defined, the map preserves addition, multiplication and 1.

Thus $\mathbb{Z}_{12} \to \mathbb{Z}_4$ is valid, but the same formula from $\mathbb{Z}_4$ to $\mathbb{Z}_{12}$ is not: $[0]_4 = [4]_4$ would have distinct images. Choosing the least nonnegative representative defines a set function but does not repair additivity. The image of $[3]_4 + [1]_4 = [0]_4$ is 0, whereas $3 + 1 = 4$ in $\mathbb{Z}_{12}$.

A valid nonunital example is

$$\psi : \mathbb{Z}_4 \to \mathbb{Z}_6, \qquad [a] \mapsto [3a].$$

It preserves addition. Multiplication is preserved since $9ab \equiv 3ab \pmod 6$, and $4 \cdot 3 \equiv 0$ ensures independence from representatives. The image $\{0, 3\}$ has its own multiplicative identity 3, rather than 1 of $\mathbb{Z}_6$. Its kernel is $\{0, 2\}$, so $\mathbb{Z}_4/\{0, 2\} \cong \{0, 3\} \cong \mathbb{Z}_2$.

14.11 Boundaries and common errors

- $(\mathbb{Z}_n, +, \cdot)$ is a ring, but nonzero classes need not be units.
- Products of nonzero elements can be zero for composite moduli.
- $\mathbb{Z} \subset \mathbb{Q}$ is a unital subring but not an ideal; $2\mathbb{Z} \subset \mathbb{Z}$ is an ideal lacking 1.
- A kernel must contain zero and absorb multiplication by the ambient ring.
- The pairwise gcd conditions must be checked before applying the coprime form of CRT.

These structures recur in RSA in Chapter 16 and finite fields in Chapter 17. A field requires additional axioms.

Transporting operations changes zero and one

On $\mathbb{Q}$, define

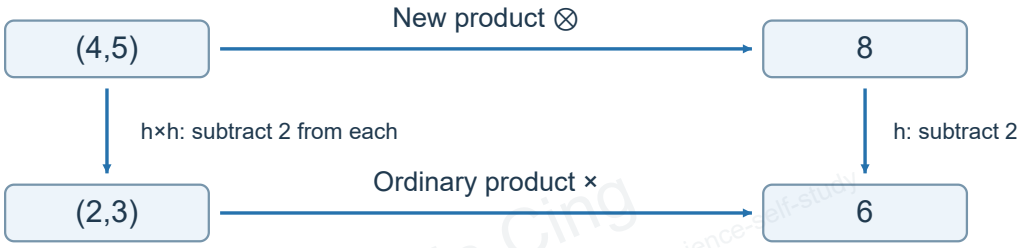
$$a \oplus b = a + b - 2, \qquad a \otimes b = (a-2)(b-2) + 2.$$

The bijection $h(a) = a - 2$ satisfies $h(a \oplus b) = h(a) + h(b)$ and $h(a \otimes b) = h(a)h(b)$. It transforms both new operations into ordinary rational operations. Transporting the ring axioms through h gives a field isomorphic to $\mathbb{Q}$.

The new additive zero is $h^{-1}(0) = 2$, and the new multiplicative identity is $h^{-1}(1) = 3$. The additive inverse of a is $4 - a$. For $a \neq 2$, its new multiplicative inverse is $2 + 1/(a-2)$. For example, the inverse of 4 is $5/2$, and $4 \otimes (5/2) = 3$, the new identity. Requiring this product to equal the ordinary numeral 1 would use the wrong operation structure.

Transporting operations changes zero and one (continued)

Transported multiplication: 4 ⊗ 5 = 8 corresponds to 2 × 3 = 6



New product then h, or h on each input then ordinary product: both paths give 6.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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